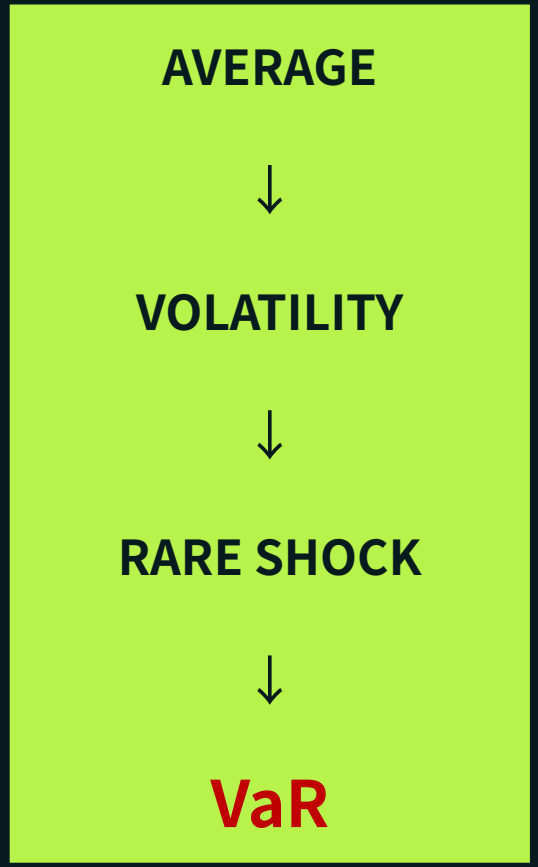
GARCH Value at Risk
A Beginner's Guide

What each symbol in one formula means



Market Risk · Beginner Session

One formula estimates how large a loss can get

VaR combines the average return, the predicted volatility and the size of a rare shock

$$\underbrace{\mathrm{VaR}_t(\alpha)}_{\text{Value at Risk}} = \underbrace{-}_{\text{flip the sign}} \left(\underbrace{\mu}_{\text{average}} + \underbrace{\sigma_t}_{\text{predicted}} \underbrace{z_\alpha}_{\text{size of a}} \right)$$

Value at Risk — a loss threshold

flip the sign — return → loss

average — return

predicted — volatility

size of a — rare bad shock

1. t — the day we forecast
4. z_t — the surprise, scaled
5. α — how rare: 1% of days

Numbers 1–7 match the slides that follow, one symbol at a time.

Key idea: when volatility rises, the bad-day loss rises too.

First, split a return into three parts

The VaR formula is built on this simple split

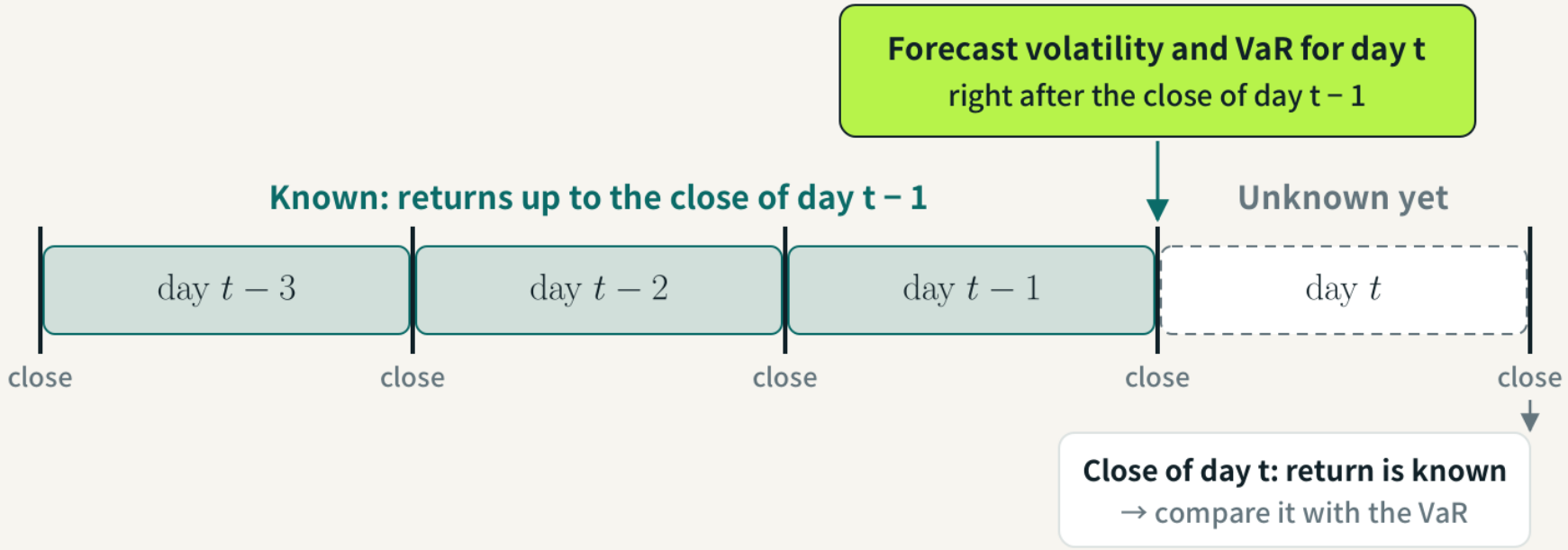
$$r_t = \underbrace{\mu}_{\text{mean}} + \underbrace{\sigma_t}_{\text{volatility}} \underbrace{Z_t}_{\text{shock}}$$

$$\text{Return} = \text{average} + \text{volatility} \times \text{surprise}$$

A bad day = average return plus a large negative surprise, scaled by volatility.

Symbol 1 · t is the day we calculate the risk for

Each block is one trading day, ending at the market close



The volatility is predicted before that day's return is known.

Symbol 2 · μ is the return we expect on average

Example: an expected daily return of 0.05%

$$\mu = 0.05\% = 0.0005$$

WHAT IT MEANS

The centre of returns

Where daily returns land on average

WHAT IT DOES NOT MEAN

Not earned every day

Some days are up, some are down

The average is a centre, not a promise.

Symbol 3 · σ is the predicted size of the daily swing

Example: GARCH predicts a daily volatility of 2%

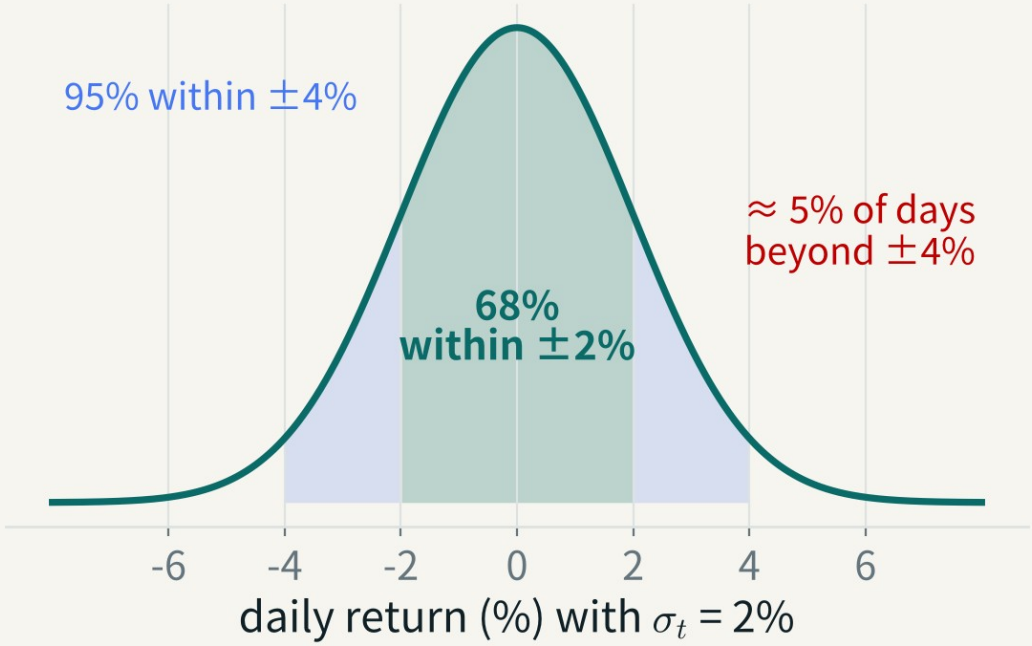
Predicted volatility

$$\sigma_t = 2\%$$

WHAT GARCH DOES

Predicts it day by day

Uses recent shocks and past volatility



2% is the typical swing, not the largest possible move.

Symbol 4 · Z is the standardised shock, also known as a z-score

How far today's return is from the average, counted in standard deviations

Definition

$$Z_t = \frac{r_t - \mu}{\sigma_t}$$

Example: a value of −2

$$Z_t = -2 \iff r_t = \mu - 2\,\sigma_t$$

HOW TO READ IT

“capital Z sub t”

A random shock on day t

OFFICIAL TERM

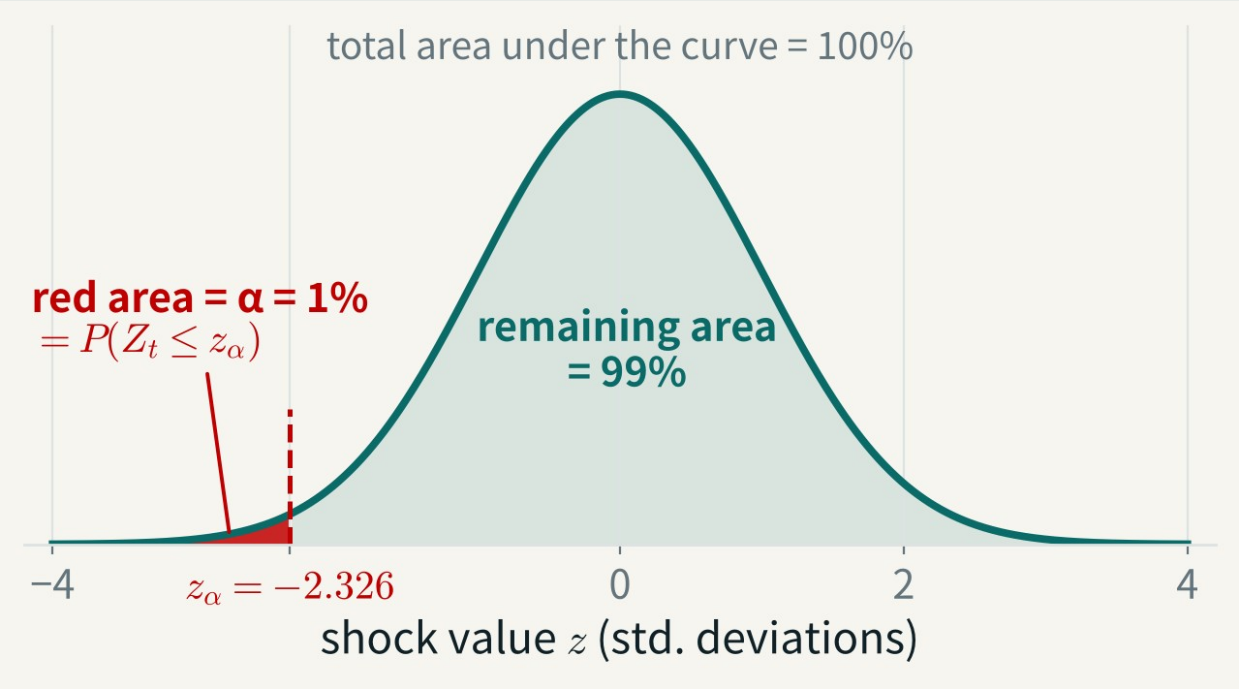
Standardised shock

Everyday name: z-score

Volatility gives the shock its size in %; Z only counts standard deviations.

Symbols 5–6 · The area left of the cut-off is the probability α

Under a probability curve, area = probability; the total area is 100%



Area left of the cut-off = α

$$P\left(Z_t \leq z_\alpha\right) = \alpha$$

α · "ALPHA"

Tail area (probability): 1% here

SMALL z SUB ALPHA

The cut-off: −2.326 for 1%

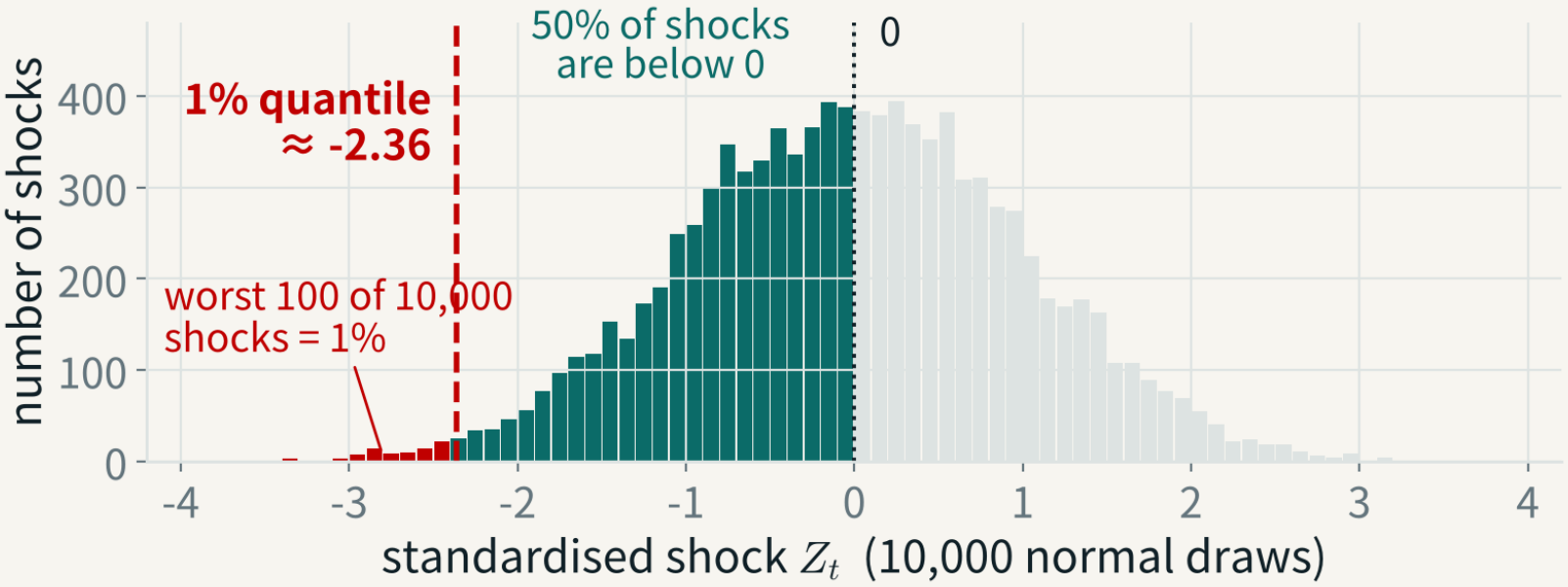
CAPITAL Z SUB t

The random shock itself

Pick the area α first; the cut-off is the point that leaves exactly that area on its left.

Symbol 6 · A quantile is the point with a given area to its left

Sort 10,000 normal shocks: 1% of them (1% of the area) lie left of the 100th smallest

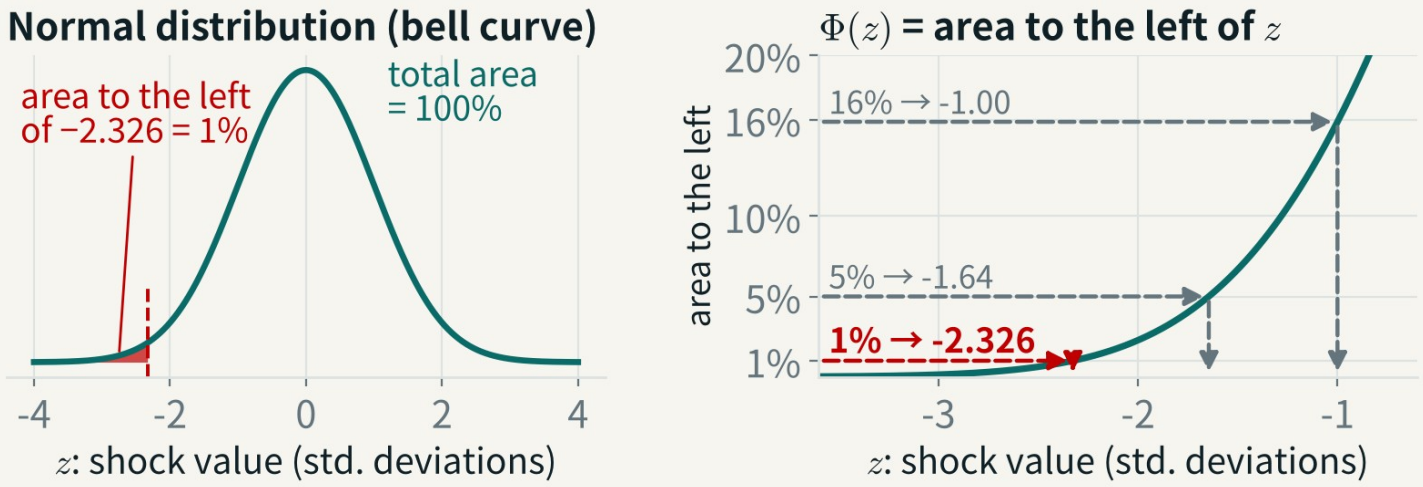


Here the 100th smallest is −2.36; with more draws it settles at −2.326.

Half of all shocks are below 0, so the 1% cut-off must be negative.

Symbol 6 · −2.326 is the point with 1% of the area to its left

Left: the bell curve. Right: Φ(z), the area under the bell curve to the left of each z



THE RIGHT-HAND CURVE

Normal CDF: area left of z

Y-AXIS

Left area = probability ≤ z

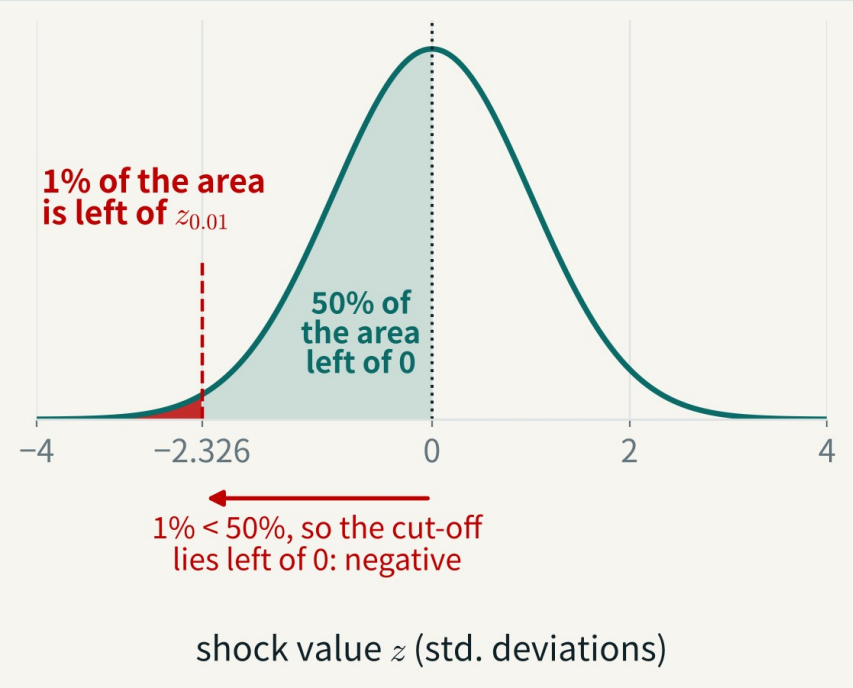
X-AXIS

Shock z, in std. deviations

Find 1% on the y-axis, go across to the curve, then down: −2.326 (Excel: =NORM.S.INV(0.01)).

Symbol 7 · How −4.602% becomes a VaR of +4.602%

Example: average 0.05%, volatility 2%, 1% tail, normal shock — in decimals, then in %



Step 1
Scale the cut-off

$$\sigma_t \, z_\alpha = 0.02 \times (-2.326) = -0.04652 = -4.652\%$$

Step 2
Add the average

$$\mu + \sigma_t \, z_\alpha = 0.0005 + (-0.04652) = -0.04602 = -4.602\%$$

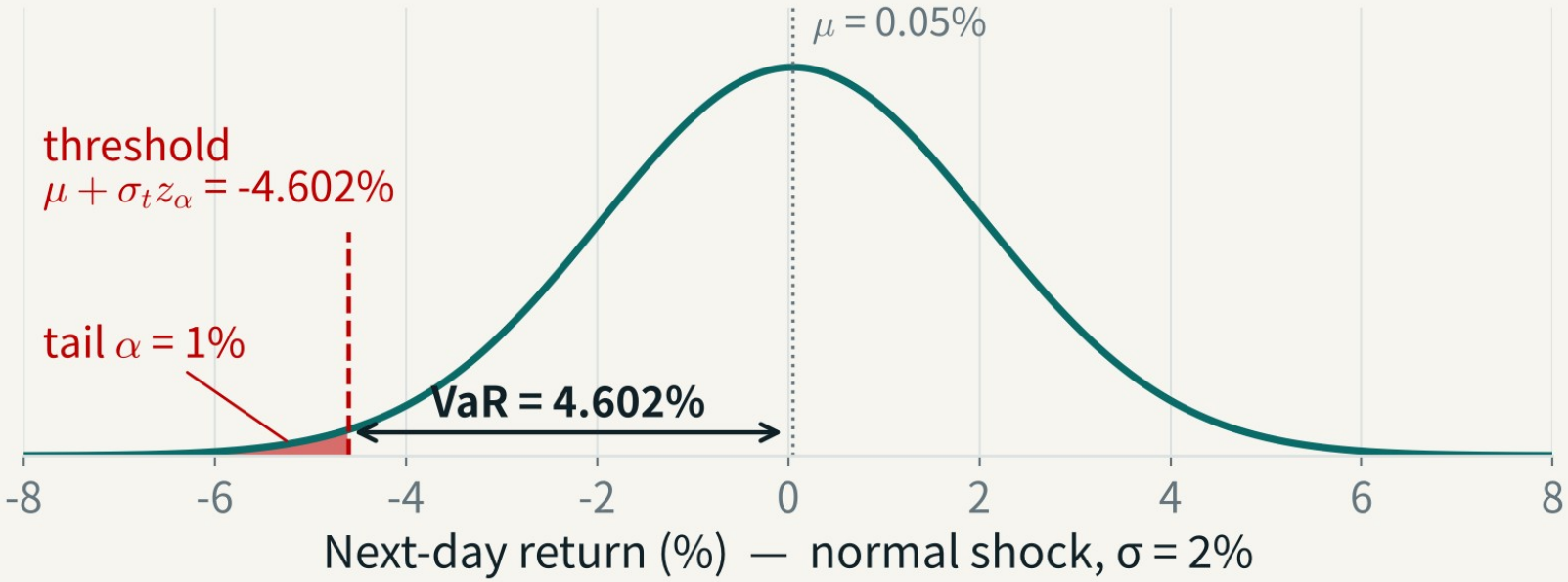
Step 3
Flip the sign

$$\mathrm{VaR}_t(0.01) = -(-4.602\%) = +4.602\%$$

Loss = −return: a return of −4.602% is a loss of 4.602%.

VaR sits at the edge of the worst 1% of days

Average 0.05%, volatility 2%, normal shock; with α = 1% this is called the 99% VaR



The red area holds the worst 1% of days; its edge is −4.602%.

VaR is the distance from zero to that edge: 4.602%.

Worked example · On 100 million KRW, VaR is about 4.6 million KRW

Step 3 flips the sign; the money amount scales the percentage by the position size

Step 3
Report as a loss

$$\boxed{\mathrm{VaR}_t(0.01) = -(-4.602\%) = 4.602\%}$$

In money terms

$$100{,}000{,}000\ \mathrm{KRW} \times 4.602\% = \boxed{4{,}602{,}000\ \mathrm{KRW}}$$

CORRECT READING

Exceeded on 1% of days

WRONG READING

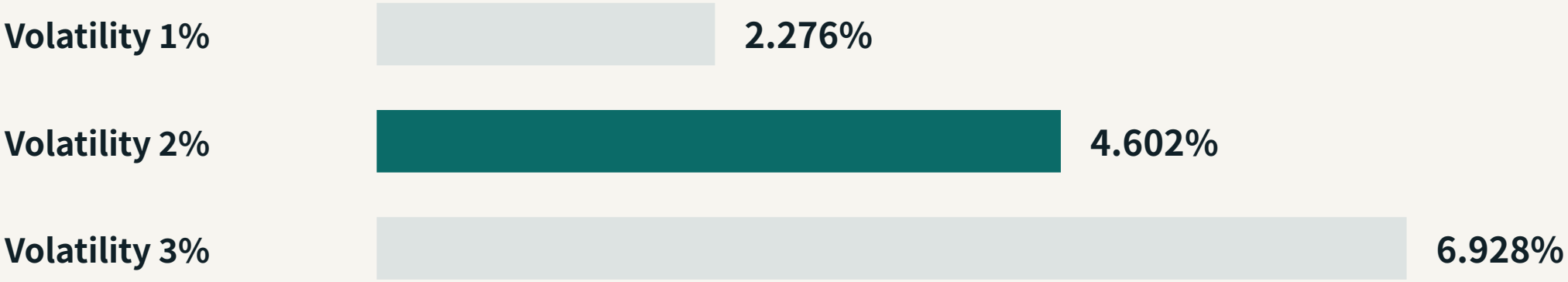
“The maximum loss”

Larger losses can and will happen — VaR says how often, not how large.

Higher volatility means a higher VaR

Same average 0.05% and 1% tail; only the GARCH volatility changes

$$\sigma_t \uparrow \Longrightarrow \mathrm{VaR}_t \uparrow$$



When GARCH sees bigger swings, tomorrow's VaR goes up.

Three things to remember

The whole formula in three sentences

GARCH

Predicts volatility

Tomorrow's swing, from recent days

DISTRIBUTION

Sets the rare shock

Normal: −2.326 for the worst 1%

READING

A threshold, not a max

Bigger losses happen on 1% of days

Average plus scaled rare shock, then flip the sign

GARCH predicts tomorrow's volatility.

The rare-shock size turns it into a bad-day return.

VaR is that loss — exceeded on 1% of days.

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※ Example: average 0.05%, volatility 2%, 1% tail, normal quantile −2.326.